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CHUCK ASSEMBLY, PLANARIZATION PROCESS, APPARATUS AND METHOD OF MANUFACTURING AN ARTICLE

Abstract

A chuck assembly for holding a plate comprises a plate holding member configured to hold the plate, the plate holding member including a central opening, a rigid member configured to hold the plate holding member, a first channel formed by the plate holding member, a second channel formed by the rigid member, a first fluid conduit in communication with the second channel, and, a second fluid conduit at least partially disposed within the second channel and in communication with the first channel.

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Background/Summary

BACKGROUND

Field of Art

[0001] The present disclosure relates to substrate processing, and more particularly, to a chuck assembly used in the planarization of surfaces in semiconductor fabrication.

Description of the Related Art

[0002] Planarization techniques are useful in fabricating semiconductor devices. For example, the process for creating a semiconductor device includes repeatedly adding and removing material to and from a substrate. This process can produce a layered substrate with an irregular height variation (i.e., topography), and as more layers are added, the substrate height variation can increase. The height variation has a negative impact on the ability to add further layers to the layered substrate. Separately, semiconductor substrates (e.g., silicon wafers) themselves are not always perfectly flat and may include an initial surface height variation (i.e., topography). One method of addressing this issue is to planarize the substrate between layering steps. Various lithographic patterning methods benefit from patterning on a planar surface. In ArFi laser-based lithography, planarization reduces the impact of depth of focus (DOF) limitations, and improves critical dimension (CD), and critical dimension uniformity. In extreme ultraviolet lithography (EUV), planarization improves feature placement and reduces the impact of DOF limitations. In nanoimprint lithography (NIL) planarization improves feature filling and CD control after pattern transfer.

[0003] A planarization technique sometimes referred to as inkjet-based adaptive planarization (IAP) involves dispensing a variable drop pattern of polymerizable material between the substrate and a superstrate, where the drop pattern varies depending on the substrate topography. A superstrate is then brought into contact with the polymerizable material after which the material is polymerized on the substrate, and the superstrate removed. Improvements in planarization techniques, including IAP techniques, are desired for improving, e.g., whole wafer processing and semiconductor device fabrication.

[0004] In current planarization methods, there may be difficulty in controlling the superstrate curvature and spread of polymerizable material near the edge of the superstrate. Furthermore, there may be difficulty in separating the superstrate from a cured film, in particular when the superstrate has a similar size to the substrate. There is a need in the art for a chuck assembly and planarization method with improved control of the superstrate curvature and improved separation from the cured film. More particularly, there is a need in the art for a chuck assembly that has improved control of the superstrate curvature at the time of spreading of formable material while also having the ability to provide consistent separation front propagation at the time of separating the superstrate from the cured material.

SUMMARY

[0005] A chuck assembly for holding a plate comprises a plate holding member configured to hold the plate, the plate holding member including a central opening, a rigid member configured to hold the plate holding member, a first channel formed by the plate holding member, a second channel formed by the rigid member, a first fluid conduit in communication with the second channel; and, a second fluid conduit at least partially disposed within the second channel and in communication with the first channel.

[0006] A planarization system comprises a chuck assembly for holding a plate, the chuck assembly comprising, a plate holding member configured to hold the plate, the plate holding member including a central opening, a rigid member configured to hold the plate holding member, a first channel formed by the plate holding member, a second channel formed by the rigid member, a first fluid conduit in communication with the second channel, and a second fluid conduit at least partially disposed within the second channel and in communication with the first channel, a substrate chuck configured to hold a substrate, a fluid dispenser configured to dispense formable material on the substrate, a positioning system configured to contact the formable material with the plate, and a curing system configured to cure the formable material under the plate so as to form cured formable material on the substrate.

[0007] A method of manufacturing an article comprises dispensing a formable material on a

substrate, retaining a plate with a plate chuck assembly, the plate chuck assembly comprising: a plate holding member configured to hold the plate, the plate holding member including a central opening, a rigid member configured to hold the plate holding member, a first channel formed by the plate holding member, a second channel formed by the rigid member, a first fluid conduit in communication with the second channel, and a second fluid conduit at least partially disposed within the second channel and in communication with the first channel, contacting the plate with the formable material dispensed on the substrate, curing the formable material with a curing source, separating the plate from the cured formable material, and processing the cured formable material to make the article.

[0008] These and other objects, features, and advantages of the present disclosure will become apparent upon reading the following detailed description of exemplary embodiments of the present disclosure, when taken in conjunction with the appended drawings, and provided claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] So that features and advantages of the present disclosure can be understood in detail, a more particular description of embodiments of the disclosure may be had by reference to the embodiments illustrated in the appended drawings. It is to be noted, however, that the appended drawings only illustrate typical embodiments of the disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

[0010] FIG. 1 is a schematic diagram illustrating an example planarization system in accordance with an aspect of the present disclosure.

[0011] FIGS. 2A to 2C illustrate a schematic cross section of an example planarization process in accordance with an aspect of the present disclosure.

[0012] FIG. 3A is a bottom view of an example chuck assembly in accordance with an aspect of the present disclosure.

[0013] FIG. 3B is a top view of the example chuck assembly of FIG. 3A.

[0014] FIG. 4A is a cross-section view of the example chuck assembly taken along line 4A-4A of FIG. 3B.

[0015] FIG. 4B is an enlarged view of portion 4B of FIG. 4A.

[0016] FIG. 4C is a perspective view of portion 4B of FIG. 4A.

[0017] FIG. 4D is an enlarged schematic view of portion 4D of FIG. 4B.

[0018] FIG. 5A is cross-section view of the example chuck assembly taken along lone 5A-5A of FIG. 3A.

[0019] FIG. 5B is an enlarged view of portion 5B of FIG. 5A.

[0020] FIG. 5C is a perspective view of portion 5B of FIG. 5A.

[0021] FIG. 5D is an exploded perspective view of the example chuck assembly.

[0022] FIG. 6A is a cross-section view of the example chuck assembly taken along line 6A-6A of FIG. 3B.

[0023] FIG. 6B is an enlarged view of portion 6B of FIG. 6A.

[0024] FIG. 6C is perspective view of the portion 6B of FIG. 6A.

[0025] FIG. 7 is an exploded bottom view of the example chuck assembly.

[0026] FIG. 8A is an enlarged view of portion 8A of FIG. 4A.

[0027] FIG. 8B is a perspective view of portion 8A of FIG. 4A.

[0028] FIG. 9 shows a flow chart of an example planarization method in accordance with aspect of the present disclosure.

[0029] FIGS. 10A to 10N show a series of schematic cross sections of the planarization method of

FIG. 9.

[0030] While the subject disclosure will now be described in detail with reference to the figures, it is done so in connection with the illustrative exemplary embodiments. It is intended that changes and modifications can be made to the described exemplary embodiments without departing from the true scope and spirit of the subject disclosure as defined by the appended claims.

DETAILED DESCRIPTION

Planarization System

[0031] FIG. 1 illustrates an example system for planarization in accordance with an aspect of the present disclosure. The planarization system **100** is used to planarize a film on a substrate **102**. The substrate **102** may be coupled to a substrate chuck **104**. The substrate chuck **104** may be but is not limited to a vacuum chuck, pin-type chuck, groove-type chuck, electrostatic chuck, electromagnetic chuck, and/or the like.

[0032] The substrate **102** and the substrate chuck **104** may be further supported by a substrate positioning stage **106**. The substrate positioning stage **106** may provide translational and/or rotational motion along one or more of the x-, y-, z-, θ -, ψ -, and ϕ -axes. The substrate positioning stage **106**, the substrate **102**, and the substrate chuck **104** may also be positioned on a base (not shown). The substrate positioning stage may be a part of a positioning system.

[0033] Spaced apart from the substrate **102** is a superstrate **108** (also referred herein as a plate) having a working surface **112** facing substrate **102**. The superstrate **108** may be formed from materials including, but not limited to, fused silica, quartz, silicon, organic polymers, siloxane polymers, borosilicate glass, fluorocarbon polymers, metal, hardened sapphire, ceramic, glass, and/or the like. In an embodiment the superstrate is readily transparent to UV light. The working surface **112** is generally of the same areal size or slightly smaller than the surface of the substrate **102**.

[0034] The superstrate **108** may be coupled to or retained by a superstrate chuck assembly **118**, which is discussed in more detail below. The superstrate chuck assembly **118** may be coupled to a planarization head **120** which is a part of the positioning system. The planarization head **120** may be movably coupled to a bridge. The planarization head **120** may include one or more actuators such as voice coil motors, piezoelectric motors, linear motor, nut and screw motor, etc., which are configured to move the superstrate chuck assembly **118** relative to the substrate **102** in at least the z-axis direction, and potentially other directions (e.g. x-, y-, θ -, ψ -, and ϕ -axis).

[0035] The planarization system **100** may further comprise a fluid dispenser **122**. The fluid dispenser **122** may also be movably coupled to the bridge. In an embodiment, the fluid dispenser **122** and the planarization head **120** share one or more of all positioning components. In an alternative embodiment, the fluid dispenser **122** and the planarization head move independently from each other. The fluid dispenser **122** may be used to deposit droplets of liquid formable material **124** (e.g., a photocurable polymerizable material) onto the substrate **102** with the volume of deposited material varying over the area of the substrate **102** based on at least in part upon its topography profile. Different fluid dispensers **122** may use different technologies to dispense formable material **124**. When the formable material **124** is jettable, ink jet type dispensers may be used to dispense the formable material. For example, thermal ink jetting, microelectromechanical systems (MEMS) based ink jetting, valve jet, and piezoelectric ink jetting are common techniques for dispensing jettable liquids.

[0036] The planarization system **100** may further comprise a curing system that includes a radiation source **126** that directs actinic energy, for example, UV radiation, along an exposure path **128**. The planarization head **120** and the substrate positioning stage **106** may be configured to position the superstrate **108** and the substrate **102** in superimposition with the exposure path **128**. The radiation source **126** sends the actinic energy along the exposure path **128** after the superstrate **108** has contacted the formable material **124**. FIG. 1 shows the exposure path **128** when the superstrate **108** is not in contact with the formable material **124**. This is done for illustrative purposes so that the

relative position of the individual components can be easily identified. An individual skilled in the art would understand that exposure path **128** would not substantially change when the superstrate **108** is brought into contact with the formable material **124**.

[0037] The planarization system **100** may further comprise a camera **136** positioned to view the spread of formable material **124** as the superstrate **108** contacts the formable material **124** during the planarization process. FIG. **1** illustrates an optical axis **138** of the field camera's imaging field. As illustrated in FIG. **1**, the planarization system **100** may include one or more optical components (dichroic mirrors, beam combiners, prisms, lenses, mirrors, etc.) which combine the actinic radiation with light to be detected by the camera **136**. The camera **136** may include one or more of a CCD, a sensor array, a line camera, and a photodetector which are configured to gather light at a wavelength that shows a contrast between regions underneath the superstrate **108** and in contact with the formable material **124** and regions underneath the superstrate **108** but not in contact with the formable material **124**. The camera **136** may be configured to provide images of the spread of formable material **124** underneath the superstrate **108**, and/or the separation of the superstrate **108** from cured formable material **124**. The camera **136** may also be configured to measure interference fringes, which change as the formable material **124** spreads between the gap between the working surface **112** and the substrate surface and as the distance between the working surface **112** and the substrate surface changes.

[0038] The planarization system **100** may be regulated, controlled, and/or directed by one or more processors **140** (controller) in communication with one or more components and/or subsystems such as the substrate chuck **104**, the substrate positioning stage **106**, the superstrate chuck assembly **118**, the planarization head **120**, the fluid dispenser **122**, the radiation source **126**, and/or the camera **136**. The processor **140** may operate based on instructions in a computer readable program stored in a non-transitory computer memory **142**. The processor **140** may be or include one or more of a CPU, MPU, GPU, ASIC, FPGA, DSP, and a general-purpose computer. The processor **140** may be a purpose-built controller or may be a general-purpose computing device that is adapted to be a controller. Examples of a non-transitory computer readable memory include but are not limited to RAM, ROM, CD, DVD, Blu-Ray, hard drive, networked attached storage (NAS), an intranet connected non-transitory computer readable storage device, and an internet connected non-transitory computer readable storage device. All of the method steps described herein may be executed by the processor **140**.

[0039] In operation, either the planarization head **120**, the substrate position stage **106**, or both vary a distance between the superstrate **108** and the substrate **102** to define a desired space (a bounded physical extent in three dimensions) that is filled with the formable material **124**. For example, the planarization head **120** may be moved toward the substrate and apply a force to the superstrate **108** such that the superstrate contacts and spreads droplets of the formable material **124** as further detailed herein.

Planarization Process

[0040] The planarization process includes steps which are shown schematically in FIGS. **2A-2C**. As illustrated in FIG. **2A**, the formable material **124** is dispensed in the form of droplets onto the substrate **102**. As discussed previously, the substrate surface has some topography which may be known based on previous processing operations or may be measured using a profilometer, AFM, SEM, or an optical surface profiler based on optical interference effect like Zygo NewView 8200. The local volume density of the deposited formable material **124** is varied depending on the substrate topography. The superstrate **108** is then positioned in contact with the formable material **124**.

[0041] FIG. **2B** illustrates a post-contact step after the superstrate **108** has been brought into full contact with the formable material **124** but before a polymerization process starts. As the superstrate **108** contacts the formable material **124**, the droplets merge to form a formable material film **144** that fills the space between the superstrate **108** and the substrate **102**. Preferably, the

filling process happens in a uniform manner without any air or gas bubbles being trapped between the superstrate **108** and the substrate **102** in order to minimize non-fill defects. The polymerization process or curing of the formable material **124** may be initiated with actinic radiation (e.g., UV radiation). For example, radiation source **126** of FIG. **1** can provide the actinic radiation causing formable material film **144** to cure, solidify, and/or cross-link, defining a cured planarized layer **146** on the substrate **102**. Alternatively, curing of the formable material film **144** can also be initiated by using heat, pressure, chemical reaction, other types of radiation, or any combination of these. Once cured, planarized layer **146** is formed, the superstrate **108** can be separated therefrom. FIG. **2c** illustrates the cured planarized layer **146** on the substrate **102** after separation of the superstrate **108**. The substrate and the cured layer may then be subjected to additional known steps and processes for device (article) fabrication, including, for example, patterning, curing, oxidation, layer formation, deposition, doping, planarization, etching, formable material removal, dicing, bonding, and packaging, and the like. The substrate may be processed to produce a plurality of articles (devices).

Spreading, Filling, and Curing Planarization Material Between Superstrate and Substrate

[0042] One scheme for minimizing entrapment of air or gas bubbles between the superstrate **108** and the substrate **102** as the formable material **124** droplets spread, merge and fill the gap between the superstrate **108** and the substrate **102** is to position the superstrate **108** such that it makes initial contact with the formable material **124** in the center of the substrate **102** with further contact then proceeding radially in a center to perimeter fashion. This requires a deflection or bowing of the superstrate **108** or substrate **102** or both to create a curvature in the superstrate **108** relative to the substrate **102**. The curvature of the superstrate **108** facilitates the expulsion of the air or gas bubbles as the formable material **124** spreads. Such a superstrate **108** profile can be obtained by, for example, applying a back pressure to the interior region of the superstrate. However, in doing so, a perimeter holding region is still required to keep the superstrate **108** retained on the superstrate chuck assembly **118**. Given that the superstrate **108** is typically of the same or similar areal dimension as the substrate **102**, if both the perimeter edges of the superstrate **108** and the substrate are **102** chucked flat during formable material **124** droplet spreading and merging, there will be no available superstrate curvature profile in this flat chucked area. This may compromise the droplet spreading and merging, which may also lead to non-fill defects in the region. To minimize non-fill defects, the superstrate curvature needs to be controlled over the full superstrate diameter during the fluid spreading process. In addition, once spreading and filling of the formable material is complete, the resultant stack of a superstrate chuck, a chucked superstrate, the formable material, substrate, and a substrate chuck can be an over-constrained system. This may cause a non-uniform planarization profile of the resultant planarized film layer. That is, in such an over-constrained system, all flatness errors or variations from the superstrate chuck, including front-back surface flatness, can be transmitted to the superstrate and impact the uniformity of the planarized film layer. Additionally, at the time of separating the superstrate from the cured film, it is desirable to achieve a consistent circumferential separation front between the superstrate and the cured film. [0043] To resolve the above issues, a superstrate chuck assembly **118** is provided as shown in FIGS. **3A** to **8B** that provides both 1) improved spreading of formable material at the time of contacting the superstrate with the formable material and 2) improved separation front propagation at the time of separating the superstrate from the cured material.

[0044] FIG. **3A** shows a bottom view of the superstrate chuck assembly **118**. FIG. **3B** shows a top view of the superstrate chuck assembly **118**. FIG. **4A** shows a cross section taken along line **4A-4A** of FIG. **3B**. FIG. **4B** shows an enlarged portion **4B** of FIG. **4A**. FIG. **4C** shows a perspective view of the enlarged portion **4B** of the FIG. **4A**. FIG. **4D** shows an enlarged schematic of portion **4D** of FIG. **4B**.

[0045] The superstrate chuck assembly **118** generally includes a rigid member **188** which may also be referred to as a support ring, a plate holding member **130** coupled with the rigid member **188**,

which is also referred herein as a member, and a light-transmitting member **150**.

[0046] The rigid member **188** preferably has a ring shape with a central opening. The rigid member **188** may be composed of an opaque material with respect to UV light or may be composed of a material that is transparent to UV light. The rigid member **188** may be composed of plastic (e.g. acrylic), glass (e.g. fused silica, borosilicate), metal (e.g. aluminum, stainless steel), or ceramic (e.g. zirconia, sapphire, alumina).

[0047] The plate holding member **130** similarly preferably has a ring shape with a central opening **132**. The plate holding member **130** may be made of a transparent material that allows UV light to pass through or may not be made of a transparent material that allows for UV light to pass through. That is the plate holding member **130** may or may not be composed of an opaque material with respect to UV light. The plate holding member **130** may be composed of a polymer (e.g., acrylic, fluoroelastomer), a glass (e.g. fused silica, borosilicate), metal (e.g. aluminum, stainless steel), or a ceramic (e.g. zirconia, sapphire, alumina). The plate holding member **130** may be configured so that the plate holding member **130** is generally flexible. However, as discussed in more detail below, portions of the plate holding member **130** can become effectively rigid when held against the rigid member **188** using a vacuum. The portion of the plate holding member **130** not held against the rigid member **188** may be considered a flexible portion. The flexible portion of the plate holding member **130** may be varied, as discussed below. However, while the portion of the plate holding member **130** that is actively flexible and actively rigid can vary, the composition of the plate holding member **130** is uniform, i.e., the same throughout. The plate holding member **130** may have a uniform thickness of 0.2 to 5 mm or 0.3 to 2 mm in an example embodiment. That is, all portions of the plate holding member **130**, whether effectively flexible or effectively rigid, may have the same thickness. A thicker material with a low elastic modulus will be similarly flexible as a thin material with high elastic modulus. The plate holding member **130** may be composed of a material having modulus of elasticity (Young's modulus) of 0.1 to 210 GPa, 50 to 150 GPa, or 60 to 100 GPa. In one example embodiment, the modulus of elasticity may be 70 GPa.

[0048] The plate holding member **130** may further have a flexural rigidity of 0.01 to 5 Pa.Math.m.sup.3, 0.1 to 4 Pa.Math.m.sup.3, 0.5 to 3 Pa.Math.m.sup.3, 1.0 to 2 Pa.Math.m.sup.3. Additionally, a ratio of the flexural rigidity of the plate holding member to the flexural rigidity of the superstrate may be 0.01:1 to 5:1, 0.05:1 to 4:1, 0.1:1 to 3:1, or 0.5:1 to 1:1, preferably less than 1:1. Equation (1) below defines the flexural rigidity D in which: H is the thickness of the superstrate **108** or the plate holding member **130**; ν is Poisson's ratio of the material of the superstrate **108** or the plate holding member **130**; and E is Young's modulus of the material of the superstrate **108** the plate holding member **130**. For example, the flexural rigidity for the superstrate may be 2.12 while the flexural rigidity of the plate holding member **130** may be 0.29, 0.68, 0.82, or 2.30 Pa.Math.m.sup.3. And the ratio of the flexural rigidity of the plate holding member **130** to the flexural rigidity of the superstrate **108** may be: 0.14:1; 0.32:1; 0.39:1; or 1.09:1.

[00001]
$$D = \frac{EH^3}{12(1-\nu^2)} \quad (1)$$

[0049] The plate holding member **130** includes a first channel **148** (FIGS. 4B, 4C, 4D, 5B, 5C, 6B, 6C) configured to hold a portion of the superstrate **108** to the underside of the plate holding member **130**. The first channel **148** may be an annular channel concentrically surrounding the central opening **132**. The first channel **148** may be located adjacent the inner edge **133** of the plate holding member **130**. The first channel **148** may be formed as a recessed portion within the thickness of plate holding member **130**. The inner edge **133** of the member is also a sidewall of the plate holding member **130** that extends from an underside surface of the plate holding member **130**.

[0050] The light-transmitting member **150** covers the central opening **132** of the plate holding member **130**. In one example embodiment, the light-transmitting member **150** is preferably transparent to UV light with high UV light transmissivity. That is, the material composition of the light-transmitting member **150** may be selected such that UV light used to cure the formable

material passes through the light-transmitting member **150**. In one example embodiment when the light-transmitting member **150** transmits UV light, the light-transmitting member may be composed of a material that transmits greater than 80% of light having a wavelength of 310-700 nm (i.e., UV light and visible light), e.g., sapphire, fused silica). In another example embodiment, the light-transmitting member need not be transparent with respect to UV light. When the light-transmitting member need not be transparent with respect to UV light, the light-transmitting member may be composed of a material that transmits greater than 80% of light having a wavelength of 400-700 nm (i.e., visible light), e.g., glass, borosilicate. That is, in the case when it is not necessary to transmit UV light, the light-transmitting member **150** should still transmit visible light. The light-transmitting member **150** is omitted from FIGS. **4A** to **6C**, **8A**, and **8B** for clarity, but is shown in FIGS. **3A**, **3B**, and **7**, and schematically shown in FIGS. **10A** to **10N**. In the illustrated example embodiment, the light-transmitting member, if shown, would rest against the innermost step **137** of the rigid member **188**. The light-transmitting member **150** may be one of: a Magnesium Fluoride (MgF.sub.2) Window; Lithium Fluoride (LiF) Window; a Barium Fluoride (BaF.sub.2); a Calcium Fluoride (CaF.sub.2) Window; a Fused Silica Window; a Polycrystalline CVD Diamond Window, a Sodium Chloride (NaCl) Window; a Potassium Bromide (Br) Window; a Sapphire Window; a Zinc Sulfide (ZnS Window); a Zinc Selenide (ZnSe) Window; a Red Thallium Bromo-Iodide (TlBr.sub.xI.sub.1-x) Window; a Gallium Arsenide (GaAs) Window; a Cadmium Telluride (CdTe) Window; a Silicon (Si) Window; a Diamond-Like Carbon (DLC) Coated Silicon Window; a Germanium (Ge) Window; a BK7 Window; a plastic window; a metal window; a ceramic window; or a glass window. The light transmitting member **150** may have one or more coatings such as anti-reflection coating or a protective coating. In an alternative embodiment, the light transmitting member **150** may be optimized to transmit infrared light.

[0051] As best seen in FIGS. **4B**, **4C** and **4D** the superstrate chuck assembly **118** may include a second channel **152** defined by the rigid member **188**. More particularly, the second channel **152** includes a first area **152a** and a second area **152b**, where the first area **152a** and the second area **152b** are continuous. Because FIGS. **4B** and **4C** are to scale, the second area **152b** is not perceivable in FIGS. **4B** and **4C**. FIG. **4D**, however, shows a schematic view of the portion **4D** in FIG. **4B**, where the sizes of the first area **152a** and the second area **152b** are exaggerated so that the second area **152b** is more clearly visible. As seen in FIG. **4D**, the first area **152a** is defined by two opposing concentric interior sidewalls **153a**, **153b** of the rigid member **188**. The first sidewall **153a** (i.e., the sidewall that is more radially inward) extends farther down relative to the first surface **153c** of the first area **152a** than the second sidewall **153b** (i.e., the sidewall that is more radially outward). The second area **152b** is defined by an edge of the second sidewall **153b** intersecting with a second surface **153e**, a third sidewall **153d**, and the second surface **153e**. The third sidewall **153d** is located more radially outward than the second sidewall **153b**. The second surface **153e** extends radially from the second sidewall **153b** to the third sidewall **153d**. The third sidewall **153d** extends down and terminates at the same relative height as the first sidewall **153a**. The second area **152b** has the same height as the third sidewall **153d**.

[0052] The height H1 is the distance in the vertical direction from the first surface **153c** serving as a reference point to the bottom of the second sidewall **153b** (i.e., the distance from the first surface **153c** to the second surface **153e**). The height H2 is the distance in the vertical direction from the first surface **153c** to the bottom of the first sidewall **153a**, which is also the distance from the first surface **153c** to the bottom of the third sidewall **153d**. As seen in FIG. **4D**, the height H2 is greater than the height H1. The height H2 may be 6.5 mm and the distance H1 may be 6.4 mm. The heights of H1 and H2 may be between 2 to 20 mm. The height H3 is the distance in the vertical direction from the second surface **153e** to the bottom of the third sidewall **153d**. The height H3 is selected based upon the material and mechanical properties of the superstrate and the mechanical material properties of the flexible portion **134**. The height H3 may be selected based on finite element simulations of the superstrate and the flexible portion during separation. If the height H3 is

too large, then the flexible portion will bend too much and the first channel will not be able to maintain vacuum with the superstrate, causing the superstrate to detach from the flex chuck and possibly be damaged. If the height H3 is too small, then the separation front will not propagate around the circumference of the substrate. In an example embodiment, the height H3 may be 5 μm to 100 μm . Thus, in an example embodiment, the first sidewall **153a** may extend 5 to 100 μm beyond the second sidewall **153b**. H2 is equal to H1 plus H3. Thus, as seen in FIG. 4D, the difference in distance between H1, H2, and H3 creates a step from the first area **152a** to the second area **152b**. This step structure provides a benefit in the step of separating the superstrate from a cured layer, which is described below. As also seen in FIG. 4D, the second channel **152**, in a state when it is enclosed, may also be considered to be defined in part by an upper surface of the plate holding member **130**. Furthermore, the first sidewall **153a** forms an innermost rigid member land of the rigid member **188** that overlaps with an innermost plate holding member land formed from the inner edge/sidewall **133** of the plate holding member **130**.

[0053] The superstrate chuck assembly **118** further includes a first fluid conduit **154** in communication with the second channel **152** for pressurizing the second channel **152** and a cavity **170** described below. The superstrate chuck assembly **118** includes a second fluid conduit **166** in communication with the first channel **148** so that the superstrate can be held by the plate holding member **130** by reducing pressure in the first channel **148**. One manner of reducing pressure in the first channel **148** is by providing a vacuum via second fluid conduit **166** and the channel **148**. The second conduit **166** may include a fitting **166a** and a tube **166b**. The second conduit **166** is best shown in FIGS. 4B and 4C. The second conduit **166** includes a first plate holding vacuum passage **172** within the fitting **166a** that ends with a fitting through hole **186**. The fitting through hole **186** is located within the bottom portion of the fitting **166a**. As best seen in FIG. 4B the bottom of the fitting **166a** does not extend beyond the confines of the second channel **152**, and more particularly, within the confines of the first area **152a**. In an embodiment, the bottom of the fitting **166a** does not extend beyond the second surface **153e**. In other words, the fitting **166a** does not extend beyond the first sidewall **153a** of the rigid member **188**. This is the case no matter the state of use of the plate holding member **130**. That is, the fitting **166a** does not move relative to the rigid member **188** regardless of whether the plate holding member **130** is fully retracted or fully extended. The different states of the plate holding member **130** are discussed below in more detail. The fitting **166a** may be sealed within the rigid member **188** via sealing members **184**. The sealing members may be gaskets or O-rings that prevent the air leakage when changing the pressure in the second channel **152**.

[0054] The fitting through hole **186** opens into the tube **166b**. The tube **166b** is best seen in FIGS. 5A to 5D. FIG. 5A is a cross-section view of the superstrate chuck assembly **118** taken along line 5A-5A of FIG. 3A. FIG. 5B is an enlarged view of portion 5B of FIG. 5A. FIG. 5C is a perspective view of portion 5B of FIG. 5A. FIG. 5D is an exploded perspective view of the superstrate chuck assembly **118**. The second conduit **166** may include components that together allow the first channel **148** to impart a vacuum onto the superstrate **108**. In the illustrated example embodiment, the second conduit **166** includes a port **168** connectable with a vacuum source (not shown). The port **168** may be connected to the vacuum source via a tube (not shown), for example. The tube **166b** may be a flexible tube having a first end **180** connected to the bottom end of the fitting **166a** and having a second end **182** connected to a second fitting attached to a channel through hole **187** formed through the plate holding member **130** and entering into the first channel **148**. That is, by being connected to both the fitting **166a** and the channel through hole **187**, the tube **166b** directs the vacuum suction downwardly into the first channel **148** via the channel through hole **187**. Thus, when fitting **166a** is connected to the vacuum source, a vacuum can be applied to first channel **148** to provide a suction force capable of coupling the area of the superstrate **108** under the first channel **148**. Similarly, a positive pressure can be applied to the first channel **148** in the same manner.

[0055] One or more additional conduits may be implemented that have the same structure as the

above-discussed second conduit **166**, where each conduit is in communication with the same first channel **148** and/or communication with a corresponding additional channel (not shown) formed in the plate holding member **130**. The additional channel or channels may be disposed concentrically around the first channel **148**. That is, the additional channel or channels may also be concentrically disposed around the central opening **132**, but may be located at a greater radial distance from the inner edge of the plate holding member **130** than the illustrated first channel **148**. In an embodiment, the inner diameter of the plate holding member **130** may be smaller and/or the first channel **148** may have additional lands. As best seen in FIG. 4A, an additional conduit **167** having the same structure as the second conduit **166** may be located at a position diametrically opposing the second conduit **166**. The additional conduits in communication with the first channel **148** assist in holding the superstrate fully to the plate holding member **130**. In another aspect, additional channels corresponding to the first channel **148** at different concentric locations allow the same superstrate chuck assembly **118** to be used with different sized superstrates.

[0056] In another embodiment, it is possible that the first channel **148** and second conduit **166** may be replaced with another mechanism for coupling the plate holding member **130** with a superstrate. For example, in place of a channel/vacuum arrangement, an electrode that applies an electrostatic force may be included. Another option is mechanical latching where a mechanical structure on the underside of the plate holding member **130** is matable with the superstrate.

[0057] As noted above, the first fluid conduit **154** is in communication with the second channel **152** to pressurize the second channel **152** and to pressurize a cavity **170**. The first fluid conduit **154** may include components that together allow the second channel **152** to be selectively positively or negatively pressurized. FIGS. 6A to 6C best show the first fluid conduit **154**. FIG. 6A is a cross-section view of the superstrate chuck assembly **118** taken along line 6A-6A of FIG. 3B. FIG. 6B is an enlarged view of portion 6B of FIG. 6A. FIG. 6C is perspective view of the portion 6B of FIG. 6A. In the illustrated example, the first fluid conduit **154** includes a first port **156** connectable with a pressurizing source (not shown). The first port **156** may be connected to the pressurizing source via a tube (not shown), for example. The first port **156** includes a first cavity pressure control passage **158** in communication with a second cavity pressure control passage **160**, where a first end **162** of the second cavity pressure control passage **160** connects with the first cavity pressure control passage **158** and a second end **164** of the second cavity pressure control passage **160** connects to the second channel **152**. More specifically, the second end **164** may open into the first area **152a** of the second channel **152**. Thus, when the first port **156** is connected to the pressurizing source, positive pressure can be applied to pressurize the second channel **152** (including both the first area **152a** and the second area **152b**) via the first fluid conduit **154**.

[0058] One or more additional fluid conduits may be implemented that have the same structure as the above-discussed first fluid conduit **154**. For example, as best seen in FIG. 6A, an additional fluid conduit **155** having the same structure as the first fluid conduit **154** may be located at a position diametrically opposing the first fluid conduit **154**. The one or more additional fluid conduits corresponding to the first fluid conduit **154** may serve the same function by opening into the second channel **152**.

[0059] FIG. 7 shows an exploded view where the rigid member **188** is shown separated from the plate holding member **130** and the light transmitting member **150**. As best seen in FIG. 7, the rigid member **188** may generally include a circular main body **190** defining an open central area **192**. The outer circumference of the rigid member **188** may be uniform. The rigid member **188** holds the light transmitting member **150** such that the light receiving member **150** overlaps the open central area **192**. The light transmitting member **150** may be secured to the rigid member **188** at the innermost step **137** with an adhesive, for example. In this manner, when the light transmitting member **150** is placed/secured to overlap the central area **192**, a cavity **170** is defined by the underside surface of the light transmitting member, and the inner surface of rigid member **188**. When a flexible portion of the plate holding member **130** below the second channel **152** extends

away from the rigid member **188**, the second channel **152** is in fluid communication with the cavity **170**.

[0060] An edge of the plate holding member **130** may be coupled to the underside surface of the rigid member **188** using a coupling member such as a screw, nut/bolt, adhesive and the like. The coupling member may preferably be located adjacent the outer edge of the rigid member **188** and adjacent the outer edge of the plate holding member **130**. When the coupling member is a screw, the coupling member preferably passes through the plate holding member **130** adjacent the outer edge of the plate holding member **130** and into the rigid member **188** adjacent the outer edge of the rigid member, such as through a plurality of receiving holes **189**. When the coupling member is an adhesive, the coupling member is preferably located between the plate holding member **130** adjacent the outer edge of the plate holding member **130** and the rigid member **188** adjacent the outer edge of the rigid member. In this manner, an upper surface of the plate holding member **130** contacts and is fixed to the underside surface of the circular main body of the rigid member **188** adjacent the outer edge of the rigid member **188** and the outer edge of the plate holding member **130**. Additional surface area of plate holding member **130** may be selectively coupled to the rigid member **188** as part of the planarization process. The manner of selectively coupling the additional surface area of the plate holding member **130** to the rigid member **188** is discussed in more detail below.

[0061] As shown in FIGS. **6A** to **6C**, a portion of the first conduit **154** and/or additional conduits serving the same function of the first conduit **154** may be contained within the rigid member **188**, and as shown in FIGS. **4A** to **5C**, a portion of the second conduit **166** and/or additional conduits serving the same function of the second conduit **166** may be contained within the rigid member **188**. More particularly, a portion of the fitting **166a** and a portion of the tube **166b** is located within the rigid member **188**. Significantly, the second channel **152**, within the rigid member **188**, provides two functions. First, the second channel **152** retains a portion of the fitting **166a** and a portion of the tube **166b**. Second, the first conduit **154** opens into the second channel **152** in order to control the pressure in the second channel **152** to extend or retract the plate holding member **130**. Thus, the second channel **152** functions to retain certain structure and also acts as a pressure control space to act upon the plate holding member **130**.

[0062] The superstrate chuck assembly **118** may further include additional fluid conduits in communication with additional channels that serve the function of selectively securing portions of the plate holding member **130** to the underside surface of the rigid member **188**. While the above-described fluid conduits communicate with the first channel **148** of the plate holding member **130** or communicate with the second channel **152**, the fluid conduits that selectively secure portions of the plate holding member **130** to the underside surface of the rigid member **188** communicate with additional annular cavities in the rigid member **188** that are open on the underside surface of the rigid member **188**.

[0063] FIGS. **8A** and **8B** show an example of a third fluid conduit **200** used for selectively securing the plate holding member **130** to the rigid member **188**. FIG. **8A** shows an enlarged portion **8A** of FIG. **4A**. FIG. **8B** shows a side perspective view of the enlarged portion **8A** of FIG. **4A**. The third fluid conduit **200** may include components that together impart a vacuum suction force onto the upper surface of the plate holding member **130** to further secure the plate holding member **130** to the underside surface of the rigid member **188**. In the illustrated example embodiment, the third fluid conduit **200** includes a first port **204** connectable with a vacuum source (not shown). The first port **204** of the third fluid conduit **200** may be connected to the vacuum source via a tube (not shown), for example. As best seen in FIGS. **8A** and **8B**, the third fluid conduit **200** includes a first member holding vacuum passage **206** connected with a second member holding vacuum passage **208**, and the second member holding vacuum passage **208** is connected with a third member holding vacuum passage **210**. As also best seen in FIGS. **8A** and **8B**, the first member holding vacuum passage **206** may be oriented vertically to direct the vacuum downwardly, the second

member holding vacuum passage **208** of the third fluid conduit **200** may be oriented horizontally to direct the vacuum radially, and the third member holding vacuum passage **210** of the third fluid conduit **200** may be oriented vertically to direct the vacuum downwardly. The third member holding vacuum passage **210** of the third fluid conduit **200** may be connected a third channel **212** having an open end facing downwardly toward the plate holding member **130**. Thus, when the first port **204** of the third fluid conduit **200** is connected to the vacuum source, and the upper side surface of the plate holding member **130** is in contact with the underside surface of the rigid member **188**, a vacuum can be applied to the third channel **212** of the rigid member **188** to secure the plate holding member **130** to the rigid member **188**, via the third fluid conduit **200**. The third channel **212** may be an annular channel located radially outward of the second channel **152**.

[0064] The same principle of the third fluid conduit **200** to control the pressure in the third channel **212** can be applied to additional channels in the rigid member **188**. For example, as shown in FIGS. **4B**, **4C**, **5B**, **5C**, **6B**, **6C**, **8A**, **8B**, a fourth channel **214** may be present in the rigid member **188**. The fourth channel **214** is also an annular channel and is located radially outward of the third channel **212**. A corresponding fourth fluid conduit **216** is included which has the same structure and function as the third fluid conduit **200**. The difference between the fourth fluid conduit **216** and the third fluid conduit **200** is only that it provides fluid communication between a vacuum source and the fourth channel **214**. Because each of the channels are at different radial locations, each channel will apply a suction force to a different annular section of the upper side surface of the plate holding member **130**. Furthermore, because each of the channels are in communication with a vacuum source via distinct flow paths, the vacuum can be independently applied to each channel. For example, if a vacuum is applied only to the third channel **212**, then the suction force will only be imparted on the portion of upper side surface of the plate holding member **130** that contacts the third channel **212**. However, if vacuum is applied to both the third channel **212** and the fourth channel **214** at the same time, then suction force will be imparted on a wider area of the upper side surface of the plate holding member **130**, i.e., the portion of the upper side surface of the plate holding member **130** that contacts the third channel **212** and the portion of the upper side surface of the plate holding member **130** that contacts the fourth channel **214**.

[0065] While the illustrated example embodiment shows two channels in addition to the second channel **152**, further channels and further corresponding fluid conduits can be included to provide finer control of how much of the member is suctioned against the rigid member **188** at a particular time in the planarization process. The number of additional channels may be chosen to provide the optimal control over how much surface area of the plate holding member **130** is suctioned underneath the rigid member **188**. For example, the number of channels may be from 1 to 10, from 3 to 7, or from 4 to 6. The channels may be of varying size. The ratio of the cross sectional area of one of the channels to the cross sectional area of another one of the channels may be from 10:1 to 1:1, from 8:1 to 4:1, or from 5:1 to 3:1. Some of the channels may have the same size and shape. The channels may have a cross section shape that is rectangular or square. The rigid member **188** may further include lands **218** between adjacent annular cavities. The lands **218** are the portion of the rigid member **188** that comes into contact with the upper surface of the plate holding member **130**.

[0066] Operation of the superstrate chuck assembly **118** as part of the planarization process will now be described with reference to FIGS. **9** and **10A** to **10N**. FIG. **9** shows a flow chart of a planarization method **900**. The method begins at step **S902**, where the substrate **102** having drops of formable material **124** dispensed thereon, is brought underneath the superstrate **108** that is coupled with the plate holding member **130** of the superstrate chuck assembly **118**. Thus, prior to performing step **S902**, the drops of formable material are dispensed onto the substrate in the manner described above. This state of step **S902** is shown in FIG. **10A**, which is a schematic cross section of the superstrate chuck assembly **118**.

[0067] Additionally, prior to forming step **S902**, the superstrate chuck assembly **118** is prepared by

applying the vacuum suction to the first channel **148** of the plate holding member **130** and contacting the first channel **148** to the upper side surface of the superstrate **108**, thereby coupling the superstrate **108** to the plate holding member **130**. In a case where there are multiple vacuum channels (e.g., 2) in the flexible portion **134** of the plate holding member **130**, in one embodiment, less than all (e.g., only one) of the vacuum channels will have a vacuum implemented during step **S902**. For example, in one embodiment, only the radially outermost first channel relative to the central opening **132** may have a vacuum imparted. However, in another embodiment, all the vacuum channels (e.g., 2) may have a vacuum implemented during step **S902**.

[0068] FIGS. **10A** through **10N** show schematic cross section view of the chuck assembly throughout the planarizing process, from forming a film layer, to curing the layer, through separation of the superstrate from a cured layer. As shown in FIG. **10A**, at the time that the substrate **102** is placed underneath the superstrate **108**, the cavity **170** is not yet pressurized with positive pressure. At the moment shown in FIG. **10A**, the pressure **P** in the second channel and the cavity are preferably equal to atmospheric pressure. Furthermore, either prior to positioning the substrate **102** under the superstrate **108** or prior to step **S904**, the vacuum suction **V** may be applied to all of the channels **212**, **214**, **152**. The vacuum imparted to the channels **212**, **214** may be strong enough to couple the plate holding member **130** to the rigid member such that the portion of the member is effectively rigid, i.e., creating a rigid portion **135**. However, the vacuum applied to the second channel **152** may be selected such that, while the inner edge **133** of the member contacts the rigid member **188**, the vacuum is not strong enough to pull the plate holding member **130** into the second channel **152**. This causes the portion of the plate holding member **130** beneath the second channel **152** to be effectively flexible, i.e., creating a flexible portion **134**.

[0069] The method may proceed to step **S904**, where the cavity **170** of the superstrate chuck assembly **118** is pressurized with positive pressure. FIG. **10B** shows a schematic cross section of the superstrate chuck assembly **118** after the cavity **170** has been pressurized. The cavity **170** may be pressurized by imparting a positive pressure **P** via the second channel **152** and the first fluid conduit **154**. The amount of pressure **P** may be selected such that it is sufficient to bow the superstrate **108** with a desired curvature, as shown in FIG. **10B**. The pressure **P** may be set to 0.1 to 10 kPa. At the same time, the vacuum suction is applied to the first channel **148**, the channel third channel **212**, and the fourth channel **214**. Thus, during step **S904**, the plate holding member **130** remains attached to the superstrate **108** via the first channel **148** and the plate holding member **130** remains attached to the rigid member **188** at the areas below the third channel **212** and the fourth channel **214**. As noted above, the portion of the plate holding member **130** that is suctioned to contact the underside of the rigid member **188** is effectively the rigid portion **135**, while the portion of the plate holding member **130** not suctioned against the underside of the rigid member **188** is effectively the flexible portion **134**. In the present context, effectively rigid means that a portion of the plate holding member that is secured to the rigid member is prevented from deflecting by more than a deflection threshold (the deflection threshold may be 0.1-10 μm depending on the planarization system) in the **z** direction towards or away from the rigid member by the forces that are available in the planarization system. The portion that is effectively rigid may be in a deflected state by more than the deflection threshold while it is secured to the rigid member. In the present context, effectively flexible means that a portion of the plate holding member can be deflected by more than the deflection threshold in the **z** direction towards and away from the rigid member and towards the substrate chuck by the forces available in the planarization system. As also shown in FIG. **10B**, because the positive pressure **P**, and the bowing of the superstrate **108**, the flexible portion **134** of the plate holding member **130** will bend/bow as well, while the rigid portion **135** does not bend. The second channel **152** may be positively pressurized to pressure **P** prior to moving the superstrate chuck assembly **118** toward the substrate **102** or as the superstrate chuck assembly **118** moves toward the substrate **102**. In the case that the pressurizing occurs while the superstrate chuck assembly **118** moves toward the substrate **102**, the target pressure should be reached prior to

the superstrate **108** coming into contact with the formable material **124**.

[0070] The method may proceed to step **S906**, where the superstrate **108** is brought into contact with the drops of formable material **124** on the substrate **102** to form a film layer **144**. FIG. **10C** shows a schematic cross section of the superstrate chuck assembly **118** just before the bowed superstrate **108** comes into contact with the drops of formable material **124**. As shown in FIG. **10C**, the positive pressure **P** is still maintained in the second channel **152** and the cavity **170**, while and the vacuum suction is still applied to the first channel **148**, the third channel **212**, and the fourth channel **214**. In an embodiment, the pressure **P** in the second channel **152** and cavity **170** is increased as the superstrate **108** conforms with the formable material **124** to maintain a desired curvature. It often requires more pressure to maintain a certain superstrate curvature as the un-conformed region of the superstrate decreases. As a contact area of the superstrate increases during step **S906** the contact area of the superstrate begins to conform to the shape of the superstrate under the contact area, while the portion of the superstrate outside the contact area is the un-conformed region in which the curvature needs to be controlled. Maintaining this curvature is important for minimizing gas trapping which can lead to non-fill defects. In an embodiment, the curvature just beyond the conformed portion (contact area) of the superstrate is controlled. In other words, the curvature of the superstrate in an annular region just outside the contact area is controlled. In an embodiment, a desired superstrate curvature profile in this annular region is controlled while formable material spreads underneath the contact area. This may require that the pressure **P** be maintained and/or increased during step **S906**. In an embodiment, the superstrate **108** is 'flat' (conforms to the shape of the substrate **102**) after the formable material has stopped spreading.

[0071] FIG. **10D** shows a schematic cross section of the superstrate chuck assembly **118** as it continues to move downwardly toward the substrate **102** to form the film **144**. As shown in FIG. **10D**, as the superstrate chuck assembly **118** continues to move the superstrate **108** downwardly, a film **144** of the formable material **124** begins to form in an area between the center of the superstrate **108** and the substrate **102**. Simultaneously with this action, the positive pressure **P** in the second channel **152** and the cavity **170** may be maintained or increased so that as the superstrate **108** is pressed against the formable material **124**, the superstrate **108** will maintain the desired curvature in the area of the superstrate that is about to conform with the formable material. Preferably, the pressure **P** is increased. That is, as seen in FIG. **10D** as compared to FIG. **10C**, the superstrate **108** has less of an arc in FIG. **10D** such that the area of the superstrate that is about to conform with the formable material maintains the desired curvature. At the same time the flexible portion **134** of the plate holding member **130** may begin to flatten along with the superstrate **108**.

[0072] FIG. **10E** shows a schematic cross section of the superstrate chuck assembly **118** at a point where the superstrate **108** is has been further pushed toward the substrate **102**. As seen in FIG. **10E**, as the superstrate **108** continues to press downwardly, the film **144** of formable material **124** spreads further along the surface of the substrate **102** toward the edges. The positive pressure **P** is further increased or maintained in the second channel **152** and the cavity **170** so as to maintain the desired curvature in the area of the substrate that is about to conform with the formable material. Preferably, the pressure **P** is further increased. Thus, as the superstrate **108** continues to press downwardly toward the substrate **102**, the superstrate **108** continues to bend to maintain the desired curvature in the area of the substrate that is about to conform with the formable material. That is, the superstrate **108** in FIG. **10E** has less of an arc than in FIG. **10D** such that the area of the superstrate that is about to conform with the formable material maintains the desired curvature. At the same time, the flexible portion **134** also continues to flatten relative to FIGS. **10C** and **10D**. That is, the flexible portion **134** is flatter in FIG. **10E** than in FIG. **10D**. The vacuum suction is still applied to the first channel **148**, the third channel **212**, and the fourth channel **214**, throughout the positions shown in FIGS. **10D** and **10E**.

[0073] FIG. **10F** shows a schematic cross section of the superstrate chuck assembly **118** at a point where the superstrate **108** has been fully pressed against the formable material **124** such that the

film **144** is fully formed. As shown in FIG. **10F**, the superstrate **108** has been pressed until it is once again flat. That is, the superstrate **108** no longer has an arc or lacks a substantial arc. Similar, the flexible portion **134** of the plate holding member **130** is flat or lacks a substantial bend. At this stage there may be a small gap between an innermost land of the rigid member **188** and the plate holding member **130** such that there is a fluid connection between the cavity **170** and the second channel. The positive pressure in the second channel **152** and the cavity **170** at this point is completely removed or open to atmosphere. The vacuum suction is still applied to the first channel **148**, the third channel **212**, and the fourth channel **214**, as the moment shown in FIG. **10E** is prior to curing and prior to the release process described below. As the forming of the film **144** is reaching completion, there may be a minor amount of positive pressure in the second channel **152** to counteract any minor amount of upward force on the portion of the plate holding member **130** below the second channel **152** caused by the pressing of the superstrate **108** against the formable film **144**. The minor amount of positive pressure at this moment would be just enough to counteract the upward force. Once the film **144** is completely formed and there is no longer any force pressing the superstrate **108** against the formable material film **144**, the pressure in the second channel **152** can be atmospheric without any positive or negative pressure applied.

[0074] As a result of the flexible portion **134** of the plate holding member **130** being coupled with the superstrate **108**, the above-described difficulty in controlling the superstrate curvature and spread of polymerizable material near the edge of the superstrate is reduced or avoided as compared to a chuck assembly lacking the flexible portion. This is because the flexible portion **134** of the plate holding member **130** has less bending stiffness than the superstrate **108** and bends with the superstrate **108** during spreading of the formable material **124**. Thus, by controlling the pressure in the second channel **152** and the cavity **170**, the flexible portion **134** is provided, and the curvature of superstrate **108** can be controlled throughout the spreading process from the center to the outer edge of the superstrate **108**. In addition to this feature of controlling the curvature of the superstrate as described above, the superstrate chuck assembly **118** also can provide consistent crack propagation during the time of separating the superstrate from the cured material. The separation aspect of the superstrate chuck assembly **118** begins with the description of FIG. **10J**.

[0075] The method may then proceed to step **S908**, where the formed film **144** located between the superstrate **108** and the substrate **102** is cured. FIG. **10G** shows a schematic cross section of the superstrate chuck assembly **118** during the curing step of step **S908** in accordance with a first example embodiment. In the first example embodiment, the curing step may be performed in the manner noted above using the curing system. The radiation source **126** may emit, for example, UV radiation that is directed through the light-transmitting member and through the superstrate **108**, each of which allow the UV radiation to pass through. In an embodiment the plate holding member **130** may be transparent to the UV radiation so that the member will not interfere with the curing process. In an embodiment the rigid member **188** may be transparent to the UV radiation so that the member will not interfere with the curing process. In another embodiment the plate holding member **130** need not be transparent with respect to UV radiation. In a case that the plate holding member **130** is opaque with respect to UV radiation, the plate holding member **130** will need to be moved relative to a multilayer structure (substrate **102**; uncured formable material film **144**; and superstrate **108**) while the uncured formable material film **144** in the multilayer combination is being cured during step **S908**. In this first example embodiment, where the UV radiation passes through the light-transmitting member, the light-transmitting member **150** may be composed of a material that transmits greater than 80% of light having a wavelength of 310-700 nm (i.e., UV light and visible light), e.g., sapphire, fused silica). After exposure to the UV radiation, the film **144** of formable material is cured, thereby forming a hardened cured layer **146**. In an embodiment, during the curing process the vacuum suction is not applied to the first channel **148**.

[0076] FIGS. **10H** and **10I** show a schematic cross section of the superstrate chuck assembly **118** during the curing step of step **S908** in accordance with a second example embodiment. In the

second example embodiment, as shown in FIG. 10H, the superstrate **108** is first released from the plate holding member **130**. Thus, at this moment the vacuum applied to the first channel **148** has been terminated. After the superstrate **108** has been released from the plate holding member **130**, the multilayer structure (combination of superstrate **108**/film **144**/substrate **102**/substrate chuck **104**) may be moved via the stage to another location. As shown in FIG. 10I, once the multilayer structure is present at the other location, the curing process can be performed. As in the first embodiment, the curing may be performed by exposing the film **144** to UV light through the superstrate **108**. However, because the multilayer structure is at another location and no longer coupled to the superstrate chuck assembly **118**, the UV light does not need to pass through the light-transmitting member **150** or through the plate holding member **130**. In this second example embodiment, where the UV radiation does not pass through the light-transmitting member, the light-transmitting member **150** may be composed of a material that transmits greater than 80% of light having a wavelength of 400-700 nm (i.e., visible light and not UV light), e.g., glass, borosilicate, and does not need to be composed of a material that transmits UV light. After the curing is complete, the multilayer structure may be brought back to the superstrate chuck assembly **118** and the superstrate **108** may be once again coupled with the plate holding member **130** by providing a vacuum to the first channel **148**.

[0077] The method may then proceed to step S910, where the superstrate **108** is separated from the cured layer **146**. In the case of the embodiment shown in FIG. 10G, the superstrate **108** has never been decoupled from the superstrate chuck assembly **118** and is therefore already positioned for separation. In the case of the embodiment shown in FIGS. 10H and 10I, after the curing is complete, as noted above, the superstrate **108** has once again been coupled with the superstrate chuck assembly **118** and is ready for separation.

[0078] The separating of the superstrate **108** from the cured layer **146** may be performed by lifting the superstrate chuck assembly **118** upwardly away from the substrate **102** along with propagating a separation front between the superstrate **108** and the cured layer **146**. The process of initiating the separation front may be performed prior to beginning to lift the superstrate chuck assembly **118**. Just prior to beginning the process of initiating the separation front, the vacuum suction applied to the third channel **212** and the fourth channel is **214** is maintained. Additionally, a vacuum is also applied to the second channel **152**. FIG. 10J shows the superstrate chuck assembly **118** just prior to initiating the separation front in an example embodiment where a vacuum is applied to first channel **148**, the second channel **152**, the third channel **212**, and the fourth channel **214**. That is, as shown in FIG. 10J, even with vacuum force applied to the second channel **152**, the portion of the plate holding member **130** below the second channel **152** is substantially flat and not pulled into the space of the second channel **152**. At the same time, the superstrate **108** is fully contacting the cured layer **146**. Thus, at the moment shown in FIG. 10J, there has not yet been any separation of the superstrate **108** from the cured layer **146**, i.e., the separation front has not yet been initiated.

[0079] FIG. 10K shows the superstrate chuck assembly **118**, at the point of initiating a separation front between the superstrate **108** and the cured layer **146**. This separation front may be initiated by raising a pin **194** that passes through an orientation feature of the substrate until it comes into contact with and pushes upwardly against the underside of the superstrate **108**. At the same time, the vacuum in the second channel **152** is maintained. As shown in FIG. 10K, once the pin **194** pushes up against the superstrate **108** to initiate the separation front, the superstrate **108** lifts from the cured layer **146**, thereby also freeing the portion of the plate holding member **130** below the second channel **152**. Because the vacuum is maintained in the second channel **152**, and because the plate holding member **130** is flexible, the portion of the plate holding member **130** under the second channel **152** is suctioned upwardly until the upper side of the plate holding member **130** contacts the second surface **153e** of the second channel **152**. Once the plate holding member **130** contacts the second surface **153e** of the second channel **152**, the portion of the plate holding member **130** under the second channel **152** is effectively rigid. In other words, because of the

vacuum force holding the portion of the plate holding member **130** against the second surface **153e**, the plate holding member **130** no longer can serve a flexible function. Initially only a small arc segment of upper side of the flexible portion opposite the pin is close to or in contact with the second surface **153e**. After this initial state and vacuum is supplied to the second channel **152**, larger portions of the upper side of the flexible portion come into contact with the second surface helping the separation front propagate around the circumference of substrate **102**. The flexible portion of plate holding member **130** is deflected in a specific manner which assists in propagating the separation front that would be lacking if the plate holding member **130** were not limited their deflection.

[0080] FIG. **10L** shows the superstrate chuck assembly **118** after the separation front initiated in FIG. **10K** has propagated around the circumference of the substrate **102**. As the vacuum is maintained in the second channel **152**, optionally along with some lifting and/or tilting of the superstrate chuck assembly **118** the separation front between the superstrate **108** and the cured layer **146** continues to propagate. The propagation first occurs circumferentially. Thus, as shown in FIG. **10K**, the separation between the superstrate **108** and the cured layer **146** has reached the opposite side of pin side in FIG. **10K**, but there has been little separation in the radial direction toward the center of the substrate **102**. As explained above, as the edges of the superstrate **108** lift from the cured layer **146**, the corresponding portion of the plate holding member **130** lifts as a result of the vacuum and comes into contact with the second surface **153e**, thereby limiting the ability of the flexible portion to be deflected beyond a specific amount. As noted above with respect to FIG. **4D**, the second surface **153e** is at the distance **H3** relative to the bottom of the sidewalls **153a**, **153d**. The distance **H3** is selected such that there is enough distance for the plate holding member **130** to lift to allow for the separation front to propagate along the circumference, while also limiting the amount of deflection of the plate holding member **130**. In other words, the distance **H3** is particularly chosen so the portion of the plate holding member **130** under the second channel **152** has enough room to lift as part of propagating the separation front along the circumference, but the distance **H3** also provides a backstop so that the plate holding member **130** and the first channel are held at a specific distance from the substrate and at a specific orientation that aids with the propagation of the separation front from the pin to around the substrate.

[0081] FIG. **10M** shows a moment after FIG. **10L** after further separation of the superstrate **108** from the cured layer **146**. In FIG. **10M**, the superstrate chuck assembly **118** has been lifted upwardly to propagate the separating from in a radial direction toward the center of the superstrate **108**. As shown in FIG. **10M**, because the portion of the plate holding member **130** below the second channel **152** is secured to the second surface by vacuum there is no additional deflection of the flexible portion. That is, the superstrate chuck assembly **118** can be continued to be lifted to propagate the separation of the superstrate **108** from the cured layer **146** without the plate holding member **130** flexing. In an embodiment, after the separation front has propagated around the circumference of the substrate **102**, the vacuum in the second channel **152** may be reduced. The flexible portion **134** of the plate holding member **130** being secured against the second surface **153e** ensures that the first channel **148** which holds the plate holding member **130** is at the desired position and orientation to ensure that the propagation of the separation front around the circumference of the substrate succeeds.

[0082] FIG. **10N** shows a moment once the separation of step **S910** has been completed just after superstrate **108** has been released from the cured layer **146**. As shown in FIG. **10N** after completing the separation, the superstrate chuck assembly **118** retains the superstrate **108** and the substrate **102** retains the cured layer **146**. The planarization process **900** can then be started again for another substrate by returning to the orientation shown in FIG. **10A**. As noted above, the planarization process **900** may be repeated many times, on the order of tens of thousands. When it is desirable to remove the superstrate **108** from the superstrate chuck assembly **118** (for example after a predetermined number of planarization processes have been completed or if some other indicator

suggests that the superstrate should be replaced), the vacuum applied to the first channel **148** may be released.

[0083] Further modifications and alternative embodiments of various aspects will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only. It is to be understood that the forms shown and described herein are to be taken as examples of embodiments. Elements and materials may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of this description.

Claims

- 1.** A chuck assembly for holding a plate, comprising: a plate holding member configured to hold the plate, the plate holding member including a central opening; a rigid member configured to hold the plate holding member; a first channel formed by the plate holding member; a second channel formed by the rigid member; a first fluid conduit in communication with the second channel; and a second fluid conduit at least partially disposed within the second channel and in communication with the first channel.
- 2.** The chuck assembly of claim 1, wherein the plate is held by the plate holding member by reducing pressure in the first channel via the second fluid conduit.
- 3.** The chuck assembly of claim 1, wherein the plate holding member is composed of a flexible material.
- 4.** The chuck assembly of claim 1, wherein a portion of the plate holding member is movable by adjusting pressure in the second channel via the first fluid conduit.
- 5.** The chuck assembly of claim 4, wherein the second channel overlaps the portion of the plate holding member.
- 6.** The chuck assembly of claim 1, further comprising: a third fluid conduit, wherein the rigid member comprises a third channel facing the plate holding member, and wherein the third fluid conduit is in communication with the third channel.
- 7.** The chuck assembly of claim 1, wherein the first fluid conduit has a first end opening into the second channel and a second end coupled with a pressure source.
- 8.** The chuck assembly of claim 1, wherein the second channel overlaps the first channel.
- 9.** The chuck assembly of claim 1, wherein the second fluid conduit has a first end opening into the first channel and a second end coupled with a pressure source.
- 10.** The chuck assembly of claim 9, wherein the second fluid conduit includes a fixture and a tube extending from the fixture, wherein the tube comprises the first end opening into the first channel, and wherein the fixture comprises the second end coupled with the pressure source.
- 11.** The chuck assembly of claim 10, wherein a portion of the fixture is positioned within the second channel.
- 12.** The chuck assembly of claim 11, wherein the portion of the fixture positioned within the second channel does not extend beyond a bottom surface of the rigid member.
- 13.** The chuck assembly of claim 10, wherein a portion of the tube is positioned within the second channel.
- 14.** The chuck assembly of claim 10, further comprising sealing members coupled with the fixture.
- 15.** The chuck assembly of claim 1, wherein the second channel is partly defined by a first sidewall extending from a first surface of the rigid member and a second sidewall extending from the first surface of the rigid member, and wherein the first sidewall extends farther from the first surface than the second sidewall.
- 16.** The chuck assembly of claim 15, wherein the first sidewall is located radially inward of the second sidewall.

- 17.** The chuck assembly of claim 16, wherein the first channel is partly defined by a third sidewall extending from the first surface of the plate holding member, and wherein the first sidewall overlaps the third sidewall.
- 18.** The chuck assembly of claim 15, wherein the first sidewall extends 5 to 100 μm beyond the second sidewall.
- 19.** A planarization system, comprising: a chuck assembly for holding a plate, the chuck assembly comprising: a plate holding member configured to hold the plate, the plate holding member including a central opening; a rigid member configured to hold the plate holding member; a first channel formed by the plate holding member; a second channel formed by the rigid member; a first fluid conduit in communication with the second channel; and a second fluid conduit at least partially disposed within the second channel and in communication with the first channel; a substrate chuck configured to hold a substrate; a fluid dispenser configured to dispense formable material on the substrate; a positioning system configured to contact the formable material with the plate; and a curing system configured to cure the formable material under the plate so as to form cured formable material on the substrate.
- 20.** A method of manufacturing an article, comprising: dispensing a formable material on a substrate; retaining a plate with a plate chuck assembly, the plate chuck assembly comprising: a plate holding member configured to hold the plate, the plate holding member including a central opening; a rigid member configured to hold the plate holding member; a first channel formed by the plate holding member; a second channel formed by the rigid member; a first fluid conduit in communication with the second channel; and a second fluid conduit at least partially disposed within the second channel and in communication with the first channel; contacting the plate with the formable material dispensed on the substrate; curing the formable material with a curing source; separating the plate from the cured formable material; and processing the cured formable material to make the article.
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